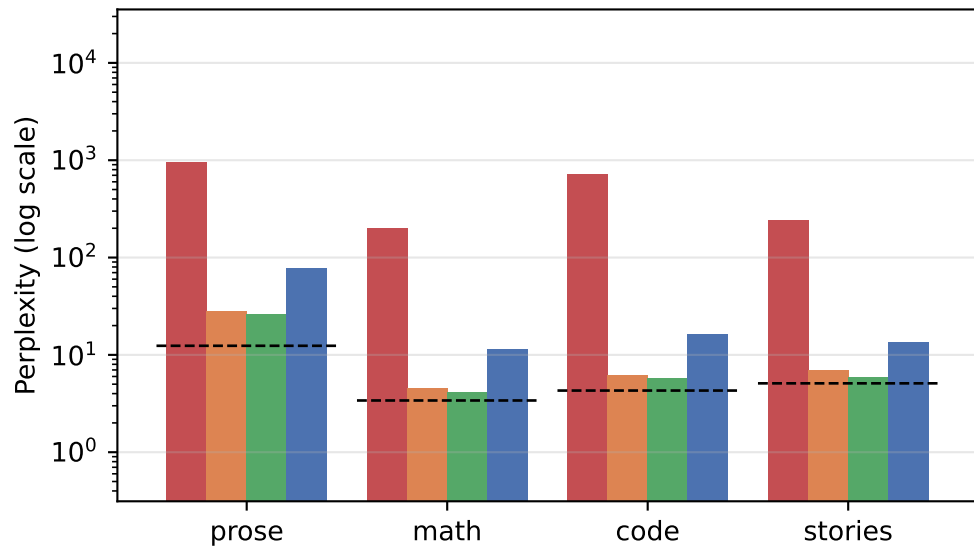


Qwen3-4B: training-free KV-sharing conversions at matched cache budget

- Unmodified baseline
- Naive mean merge (GQA-style)
- Procrustes align + merge (Jin et al. style)
- Multi-reference reconstruction (Shaikh et al. style)
- Single-leader + ridge adapters (ours)

2 × compression beyond GQA ($G = 4$)



4 × compression beyond GQA ($G = 2$)

